

Avatar++: Fast and Pose-Controllable 3D Human Avatar Generation from a Single Image

Seonghee Han^{1, #} Minchang Chung^{1, #} Gyeongsu Cho¹ Kyungdon Joo^{1, *} Taehwan Kim^{1, *}

¹Ulsan National Institute of Science and Technology (UNIST), Republic of Korea

{seonghee, minchang, threedv, kyungdon, taehwankim}@unist.ac.kr ^{*}†

Abstract

We introduce Avatar++ as an optimization-free pipeline that converts a single frontal photograph into a 3D representation in a single forward pass, taking less than 15 seconds. Single-image avatar reconstruction is difficult due to the complex structure of the human body and fine facial details, while most existing models employ Score Distillation Sampling [5, 25] or iterative refinement [4]. Our model introduces a multi-view Diffusion Transformer (DiT) with dual facial and global visual embeddings, which generates viewpoint-aligned images that preserve the subject’s facial identity. We introduce an attention mechanism that propagates information from the input image during sampling to enhance visual quality. We additionally give guidance on the pose to generate either a canonical pose (e.g., T-pose or A-pose) or replicate the pose from the input image using OpenPose [1]. These contributions position Avatar++ as a unified and efficient framework for generating identity-consistent and pose-controllable 3D human avatars from a single image. The proposed model achieves state-of-the-art performance on Thuman2.0 and RenderPeople benchmarks across all evaluation metrics, while delivering a 5× faster inference time than the fastest existing method.

1. Introduction

As virtual content becomes more widespread, interest in automated 3D avatar generation has grown, especially for mixed reality and virtual communication. Manually creating a realistic human avatar requires multi-view capture and manual modeling, which is slow and costly. To address this, generative methods that synthesize 3D avatars from limited input, such as a single image, have gained significant attention [5, 14, 15, 22, 25].

In this work, we propose Avatar++, a fast, unified, and optimization-free pipeline that generates a pose-controllable, animation-ready 3D avatar from a single image in under 15 seconds. By leveraging identity-aware em-

beddings and pose-guided multi-view generation, Avatar++ bridges the gap between personalization and efficient 3D avatar synthesis, achieving state-of-the-art multi-view generation using consumer-grade GPUs (Figure 1).

We first extract semantic facial identity embedding via ArcFace [2] and contextual visual features (clothing, pose, and overall appearance) using CLIP vision encoder which faithfully reflects both identity and visual context. Next, we propose a modified cross-attention mechanism that temporarily injects reference guidance into the query. Unlike conventional approaches that condition keys and values, we concatenate the diffusion and reference latents, both representing the same subject, into the query, enhancing identity awareness without altering the generative path. Lastly, we employ a pose-guided multi-view generation using ControlNet [24] to provide explicit human body anatomy. By conditioning on 2D openpose [1], our method overcomes for this limitation by injecting precise structural cues. By combining our multi-view synthesis model and a fine-tuned Large Gaussian Model (LGM) [18], the two-stage pipeline produces high-quality, full-body avatars while significantly reducing computational demands.

2. Related Work

2.1. Multi-view Diffusion Models

A foundational contribution to the multi-view diffusion field is zero-1-to-3, which first proposed using camera viewpoints as control conditions for image diffusion models to achieve novel view synthesis. However, it often produces inconsistencies across views due to the stochastic nature of diffusion models. Subsequent approaches, such as Zero123++ [16], shifted to an all-view-at-once generation to mitigate the inconsistency. Many prior methods [8], [6], [17], [19], [9] suffered from view inconsistency and relied on SMPL [10]. Avatar++ leverages an Identity-Aware Diffusion Transformer, a Pose-ControlNet for canonical poses to generate high-quality and cross-view consistent multi-views from a single frontal photo.

[#] Equal contribution.

^{†*} Corresponding authors.



Figure 1. comparison of in-the-wild avatar generation. Each pair shows our result (left) and Human3Diffusion [22] (right).

2.2. Single image human reconstruction

Human reconstruction from a single image typically follows either parametric or implicit approaches. Parametric methods (e.g., SMPL [10], SMPL-X [12]) predict mesh parameters but struggle with loose clothing due to fixed topology. Implicit methods (e.g., PiFU [14], SIFU [25]) offer flexible geometry via continuous fields, though they often require side-view cues or precise alignment. Recent diffusion-based methods like TeCH [5] synthesize occluded regions but rely on SMPL-X and heavy optimization. While 3D Gaussian Splatting (3DGS) methods like HumanSplat [11] achieve efficiency and quality via SMPL priors, they struggle to generalize across diverse body types and clothing. Our approach replaces SMPL with a diffusion Transformer that learns directly from images, enabling accurate, versatile multi-view synthesis without pose-specific dependencies.

3. Method

Our proposed method, as illustrated in Figure 2, reconstructs a 3D human avatar from one frontal image. Our approach consists of a two-stage process: multi-view diffusion and Gaussian reconstruction, achieving high-quality reconstructions. The first stage generates multi-view images of the human subject from a single input by leveraging the 2D diffusion transformer, while the second stage leverages these synthesized views to synthesize a 3D avatar using Gaussian splatting.

3.1. Image conditioning mechanism

To generate consistent multi-view images from a single front view, our model employs two complementary mechanisms: (1) a multi-embedding strategy to preserve facial and outfit features, and (2) an attention injection module that enriches cross-attention with reference image context. During the conditional forward pass, encoded reference features are stored in memory and later concatenated with current condition-

ing inputs during generation. Together, these mechanisms ensure identity- and appearance-preserving synthesis.

3.2. Pose Guidance with ControlNet for better Avatar Generation

To address occlusion and incomplete views in avatar synthesis, we generate multi-view images under controllable canonical poses that maximize body part visibility. This ensures structural consistency and pose-awareness, aiding animation and appearance fidelity. Pose ControlNet is trained on the MVHumanNet [20] dataset and the Renderpeople [13] dataset. Our Renderpeople subset contains rigged human models originally in canonical poses, which we retarget to cover a wide range of diverse body poses. MVHumanNet provides multi-frame sequences of the same subject with varying poses.

3.3. Large Multi-View Gaussian Model for Avatar Generation

To further enhance our avatar reconstruction quality, we integrate the Large Multi-View Gaussian Model (LGM) [18] in our second-stage pipeline. LGM efficiently generates high-resolution 3D Gaussian representations from multi-view images synthesized by our diffusion transformer. Specifically, we use its asymmetric U-Net architecture to predict compact and expressive Gaussian features from viewpoint-aligned images. These features are then fused via 3D gaussian splatting, creating a detailed and 3D avatar with significantly improved fidelity and structural accuracy.

4. Experiment

4.1. Implementation details

We train our model on 14,632 3D human samples from 2K2K [3], THuman2.0/2.1 [23], Renderpeople [13], and MVHumanNet [20]. All assets are rendered into four standardized views (every 90°) at 4096×4096 resolution, then

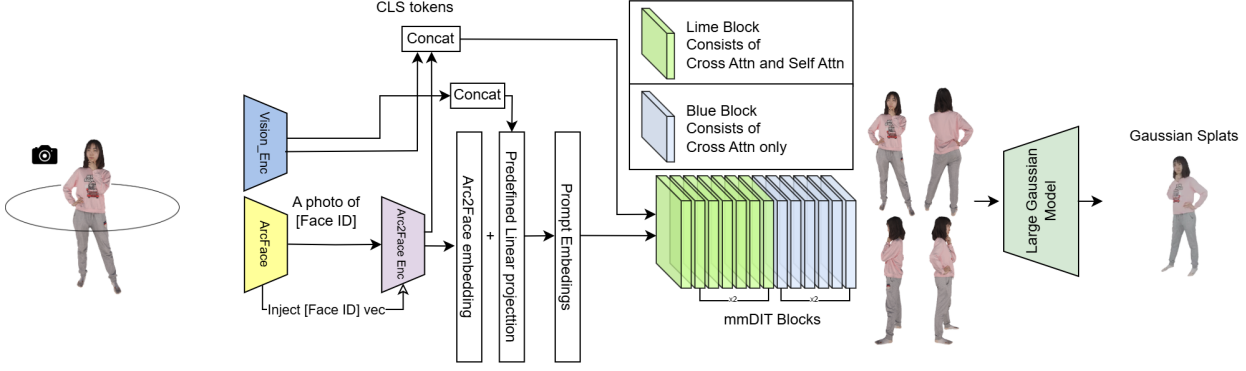


Figure 2. The overall architecture of our proposed model. The symbol + denotes concatenation. The predefined linear projection is a fixed mapping that projects the pooled output of the vision encoder into the prompt embedding space.

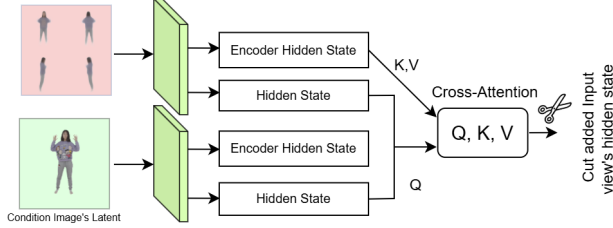


Figure 3. Hidden state injection during Cross-Attention for injecting conditioning image's information.

	Inference Time (on RTX 3090)
TeCH	10428s
SIFU	80.81s
SIFU Texture Refinement	349s + 80.81s = 429.81s
MagicMan	497.6 s
Human3Diffusion GS	68.96s
Ours(multi-view)	11.28 s
Ours(GS)	11.28 s + 2.52 s = 13.79 s

Table 1. Inference time measured on RTX 3090

resized to 512×512. We extract ArcFace embeddings from each view for detailed facial identity.

We trained our model on four Nvidia A100 GPUs (80GB) for approximately 560000 iterations and Pose controlnet for approximately 280000 iterations. LGM[18] was then finetuned on a single Nvidia A6000 GPU using the combined 2K2K, Renderpeople, and Thuman 2.1 dataset.

	Thuman2.0		
	PSNR	SSIM	LPIPS
TeCH ⁺	17.004 / 16.496	0.852 / 0.843	0.137 / 0.146
SIFU (W/O Texture Refinement)	18.33 / 18.01	0.878 / 0.875	0.115 / 0.115
SIFU (Texture Refinement) ⁺	19.699 / 19.482	0.872 / 0.870	0.118 / 0.123
MagicMan ^o	19.186 / 18.07	0.863 / 0.853	0.122 / 0.13
Human-3Diffusion (GS)	20.146 / 19.729	0.884 / 0.880	0.108 / 0.106
Ours (GS)	21.086* / 20.634*	0.896* / 0.891*	0.097* / 0.101*

Table 2. Quantitative results on the THuman2.0 dataset. ‘+’ indicates optimization-based models and ‘o’ indicates iterative methods.

4.2. Evaluation

We evaluated our model against existing baselines to measure overall performance and conducted further ablation

	RenderPeople		
	PSNR	SSIM	LPIPS
TeCH ⁺	18.683 / 18.212	0.870 / 0.862	0.119 / 0.127
SIFU (W/O Texture Refinement)	20.389 / 19.975	0.894 / 0.888	0.096 / 0.102
SIFU (Texture Refinement) ⁺	23.701 / 23.078	0.894 / 0.888	0.094 / 0.101
MagicMan ^o	24.224 / 23.535	0.890 / 0.889	0.090 / 0.098
Human-3Diffusion (GS)	24.262 / 23.744	0.900 / 0.894	0.091 / 0.096
Ours (GS)	24.637* / 24.200*	0.904* / 0.900*	0.088* / 0.092*

Table 3. Quantitative results on the RenderPeople dataset. ‘+’ indicates optimization-based models and ‘o’ indicates iterative methods.

studies to analyze the contribution of each component. The baselines include TeCH [5], SIFU [25], and MagicMan [4], which are all optimization-based methods relying on iterative refinement. In contrast, Human-3Diffusion [22] and our model both employ a single feed-forward pass for faster inference. Our model requires 2D pose annotations (from OpenPose [1]) as input. During inference, we extract 2D poses for the remaining four views by projecting the reconstructed ECON [21] mesh into those views and applying OpenPose on the projected images.

4.2.1. Quantitative Evaluation

We evaluate on 21 THuman2.0 [23] and 10 RenderPeople subjects using two view sets per subject: four canonical views (frontal, right, rear, left) and twenty dense views sampled every 18°. Evaluation metrics include PSNR, SSIM, and LPIPS, averaged over each view set. Tables 2–5 report results in the format: 4-view / 20-view. Tables 2 and 3 show that our method consistently outperforms TeCH, SIFU, MagicMan, and Human-3Diffusion on both datasets. These improvements validate the effectiveness of our identity- and pose-conditioned multi-view diffusion framework, which achieves high fidelity with a single forward pass—without relying on iterative optimization.

As Table 1 shows, on an RTX 3090, our method completes inference in 13.8s versus 69s (Human 3Diffusion), 498s (MagicMan), 429s (SIFU w / refinement), and 10428s (TeCH), achieving a 5–900× speed-up.

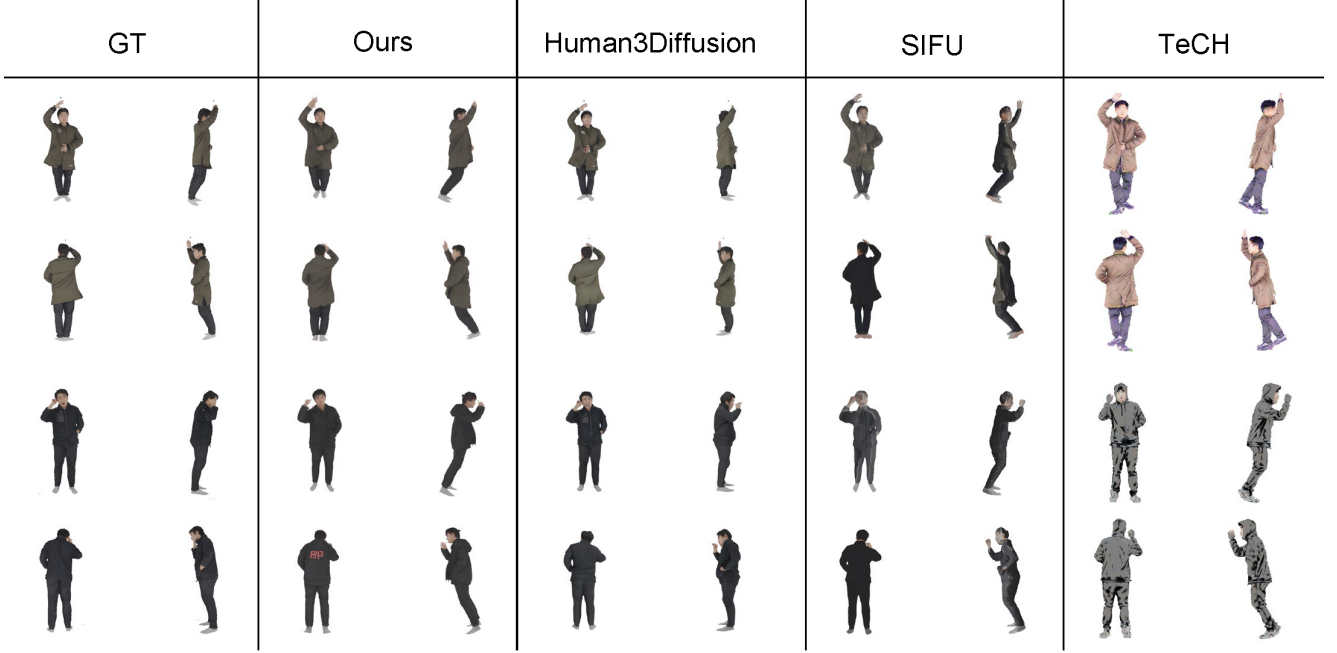


Figure 4. Qualitative results on THuman2.0 dataset. Our method captures appearance faithfully. Please zoom in for a detailed inspection.

4.2.2. Qualitative Evaluation

Figure 4 presents qualitative comparisons on THuman2.0. Human3Diffusion suffers from blurry or distorted facial features due to its low-resolution (256×256) backbone and lack of explicit facial guidance. Due to its separate optimization pipeline, SIFU produces input-aligned frontal textures but fails to infer coherent textures in unseen regions. TeCH relies on BLIP [7] for captioning, which often generates inaccurate annotations (e.g., misidentifying ethnicity), leading to inconsistent identity cues. In contrast, our method preserves fine-grained details and achieves coherent full-body synthesis through identity-aware embeddings and pose-guided multi-view generation.

4.3. Ablation studies

As part of our ablation studies, we evaluate structurally modified versions of our model by removing face embeddings and attention injection to assess their impact.

Table 4 and Table 5 show that both components contribute significantly to the final performance and demonstrate that both face embeddings and attention injection are essential for achieving high-fidelity, identity-consistent multi-view image synthesis, contributing to sharper textures, better structural similarity, and perceptual quality.

5. Conclusion

Avatar++ provides a streamlined solution for creating high-quality, animation-ready 3D avatars from a single image. Leveraging dual embeddings (ArcFace and CLIP), attention injection, and pose-guided multiview synthesis, it

	Thuman2.0		
	PSNR	SSIM	LPIPS
W/O attn injection	18.894 / 18.697	0.850 / 0.862	0.133 / 0.126
W/O Face embeddings	18.789 / 18.542	0.866 / 0.862	0.124 / 0.128
Ours (GS)	21.086 / 20.634	0.896 / 0.891	0.097 / 0.101

Table 4. Ablation studies on THuman2.0 dataset

	RenderPeople		
	PSNR	SSIM	LPIPS
W/O attn injection	20.976 / 20.824	0.884 / 0.882	0.109 / 0.112
W/O Face embeddings	22.000 / 21.770	0.887 / 0.884	0.106 / 0.109
Ours (GS)	24.637 / 24.200	0.904 / 0.900	0.088 / 0.092

Table 5. Ablation studies on the RenderPeople dataset

achieves superior fidelity and speed 5× faster than prior methods—while outperforming baselines on THuman2.0 and RenderPeople. This makes Avatar++ well-suited for applications in VR, gaming, and real-time communication. However, Avatar++ has limitations on Pose estimation. To synthesize multi-view images with the same pose as the input, our method relies on the external ECON framework for 3D pose estimation. However, this dependency can introduce inaccuracies during inference, especially under challenging articulations or occlusions. Also, its performance is constrained by limited and biased training data. THuman2.1 and MVHumanNet datasets are heavily biased toward East Asian subjects, resulting in possible racial bias. Future work will therefore focus on developing more robust pose predictors to address ECON-related error. By overcoming these limitations, we expect Avatar++ to evolve into an even more faithful and equitable solution for real-time avatar creation across all users.

References

- [1] Z. Cao, G. Hidalgo Martinez, T. Simon, S. Wei, and Y. A. Sheikh. 2019. OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2019).
- [2] Jiankang Deng, Jia Guo, Xue Niannan, and Stefanos Zafeiriou. 2019. ArcFace: Additive Angular Margin Loss for Deep Face Recognition. In *CVPR*.
- [3] Sang-Hun Han, Min-Gyu Park, Ju Hong Yoon, Ju-Mi Kang, Young-Jae Park, and Hae-Gon Jeon. 2023. High-fidelity 3d human digitization from single 2k resolution images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12869–12879.
- [4] Xu He, Zhiyong Wu, Xiaoyu Li, Di Kang, Chaopeng Zhang, Jiangnan Ye, Liyang Chen, Xiangjun Gao, Han Zhang, and Haolin Zhuang. 2025. MagicMan: Generative Novel View Synthesis of Humans with 3D-Aware Diffusion and Iterative Refinement. *Proceedings of the AAAI Conference on Artificial Intelligence* 39, 3 (Apr. 2025), 3437–3445. doi:10.1609/aaai.v39i3.32356
- [5] Yangyi Huang, Hongwei Yi, Yuliang Xiu, Tingting Liao, Jiaxiang Tang, Deng Cai, and Justus Thies. 2024. Tech: Text-guided reconstruction of lifelike clothed humans. In *2024 International Conference on 3D Vision (3DV)*. IEEE, 1531–1542.
- [6] Zehuan Huang, Hao Wen, Juntao Dong, Yaohui Wang, Yangguang Li, Xinyuan Chen, Yan-Pei Cao, Ding Liang, Yu Qiao, Bo Dai, and Lu Sheng. 2024. EpiDiff: Enhancing Multi-View Synthesis via Localized Epipolar-Constrained Diffusion. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE Computer Society, Los Alamitos, CA, USA, 9784–9794. doi:10.1109/CVPR52733.2024.00934
- [7] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. 2022. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*. PMLR, 12888–12900.
- [8] Peng Li, Yuan Liu, Xiaoxiao Long, Feihu Zhang, Cheng Lin, Mengfei Li, Xingqun Qi, Shanghang Zhang, Wenhan Luo, Ping Tan, et al. 2024. Era3D: High-Resolution Multiview Diffusion using Efficient Row-wise Attention. *arXiv preprint arXiv:2405.11616* (2024).
- [9] Xiaoxiao Long, Yuan-Chen Guo, Cheng Lin, Yuan Liu, Zhiyang Dou, Lingjie Liu, Yuexin Ma, Song-Hai Zhang, Marc Habermann, Christian Theobalt, et al. 2023. Wonder3D: Single Image to 3D using Cross-Domain Diffusion. *arXiv preprint arXiv:2310.15008* (2023).
- [10] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. 2015. SMPL: A Skinned Multi-Person Linear Model. *ACM Trans. Graphics (Proc. SIGGRAPH Asia)* 34, 6 (Oct. 2015), 248:1–248:16.
- [11] Panwang Pan, Zhuo Su, Chenguo Lin, Zhen Fan, Yongjie Zhang, Zeming Li, Tingting Shen, Yadong Mu, and Yebin Liu. 2024. Humansplat: Generalizable single-image human gaussian splatting with structure priors. *Advances in Neural Information Processing Systems* 37 (2024), 74383–74410.
- [12] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. 2019. Expressive Body Capture: 3D Hands, Face, and Body from a Single Image. In *Proceedings IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*.
- [13] Renderpeople. 2024. Renderpeople 3D Human Model Dataset. <https://renderpeople.com/> Accessed: 2025-05-20.
- [14] Shunsuke Saito, Zeng Huang, Ryota Natsume, Shigeo Morishima, Angjoo Kanazawa, and Hao Li. 2019. Pifu: Pixel-aligned implicit function for high-resolution clothed human digitization. In *Proceedings of the IEEE/CVF international conference on computer vision*. 2304–2314.
- [15] Shunsuke Saito, Tomas Simon, Jason Saragih, and Hanbyul Joo. 2020. Pifuhd: Multi-level pixel-aligned implicit function for high-resolution 3d human digitization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 84–93.
- [16] Ruoxi Shi, Hansheng Chen, Zhuoyang Zhang, Minghua Liu, Chao Xu, Xinyue Wei, Linghao Chen, Chong Zeng, and Hao Su. 2023. Zero123++: a Single Image to Consistent Multi-view Diffusion Base Model. *arXiv:2310.15110 [cs.CV]*
- [17] Yichun Shi, Peng Wang, Jianglong Ye, Mai Long, Kejie Li, and Xiao Yang. 2024. MVDream: Multi-view Diffusion for 3D Generation. In *International Conference on Learning Representations (ICLR)*.
- [18] Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. 2024. LGM: Large Multi-view Gaussian Model for High-Resolution 3D Content Creation. In *ECCV (4)*. 1–18. https://doi.org/10.1007/978-3-031-73235-5_1
- [19] Peng Wang and Yichun Shi. 2023. ImageDream: Image-Prompt Multi-view Diffusion for 3D Generation. *arXiv:2312.02201 [cs.CV]* <https://arxiv.org/abs/2312.02201>
- [20] Zhangyang Xiong, Chenghong Li, Kenkun Liu, Hongjie Liao, Jianqiao Hu, Junyi Zhu, Shuliang Ning,

- Lingteng Qiu, Chongjie Wang, Shijie Wang, et al. 2024. Mvhumannet: A large-scale dataset of multi-view daily dressing human captures. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 19801–19811.
- [21] Yuliang Xiu, Jinlong Yang, Xu Cao, Dimitrios Tzionas, and Michael J. Black. 2023. ECON: Explicit Clothed humans Optimized via Normal integration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [22] Yuxuan Xue, Xianghui Xie, Riccardo Marin, and Gerard. Pons-Moll. 2024. Human 3Diffusion: Realistic Avatar Creation via Explicit 3D Consistent Diffusion Models, In NIPS. *NeurIPS 2024*.
- [23] Tao Yu, Zerong Zheng, Kaiwen Guo, Pengpeng Liu, Qionghai Dai, and Yebin Liu. 2021. Function4D: Real-time Human Volumetric Capture from Very Sparse Consumer RGBD Sensors. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR2021)*.
- [24] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2023. Adding Conditional Control to Text-to-Image Diffusion Models. arXiv:2302.05543 [cs.CV] <https://arxiv.org/abs/2302.05543>
- [25] Zechuan Zhang, Zongxin Yang, and Yi Yang. 2024. SIFU: Side-view Conditioned Implicit Function for Real-world Usable Clothed Human Reconstruction. arXiv:2312.06704 [cs.CV] <https://arxiv.org/abs/2312.06704>